

Artist-Conditional Lyric Generation with QLoRA Adapter Blending

CENG 467 — Natural Language Understanding & Generation

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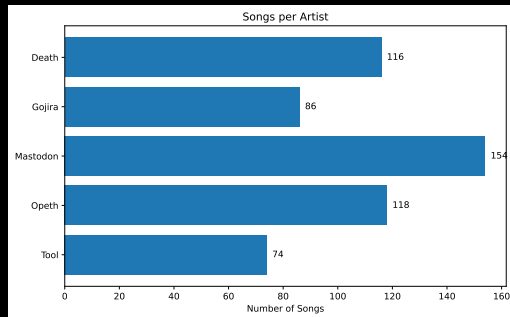
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The data: five same-genre metal artists

Artist	Songs	Style
Mastodon	154	sludge / prog
Death	116	technical / philos.
Opeth	118	poetic / melancholic
Gojira	86	environmental
Tool	74	cryptic / progressive

Same genre on purpose: the model must learn *artist*, not genre.

Note the imbalance — it matters later.



Baseline 1 — pretrained gemma-3n-E4B: junk

[User 0001] This is my band/artist page where you enter your range of audio, video -- preset unavailable for your region. To read lyrics from web, click back in your browser...

[User 0001] [ame=http://youtube.com/...]YouTube - Gero booyy[/video][ame] ...you are wanting to post a video, hide it from devices. Song Title 0 replies Added Friday, July 24, 2013

[User 0001] GOJIRA! [User 0002] I'll be going down to the woods today
Noooooooooooo!!!!!! [User 0003] In every dark place that this Gojira go..

The base model is a *text predictor*, not a lyricist: it continues scraped forum/HTML noise. No prompt fixes this.

Baseline 2 — instruct gemma-3n-E4B-it: fluent, but one voice

Gojira: *Beneath the obsidian shroud, where silence breeds its dread reign*

Mastodon: *Beneath the obsidian canopy, where fungal tides now creep and churn*

Opeth: *Beneath the glacial shroud, a silence reigns supreme*

Death: *Visceral rot descends on brittle bone*

Tool: *The aperture narrows on a familiar hum*

- ▶ **Intra-artist:** all 10 Gojira samples open “Beneath the obsidian/basalt shroud” — near-zero variance.
- ▶ **Inter-artist:** Gojira $\approx$ Mastodon $\approx$ Opeth all write “Beneath the obsidian canopy.”

Two baselines, two failures: base = incoherent junk; instruct = coherent but *the same poem for every band*. Fluency is not style.

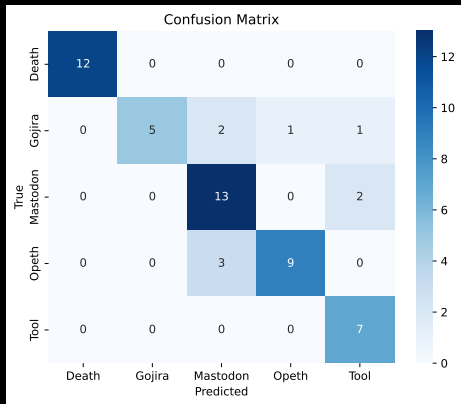
So we fine-tune: one QLoRA adapter per artist

- ▶ Base: **Gemma 3n E4B** ($\sim 4\text{B}$), QLoRA 4-bit NF4
- ▶ One **LoRA r8** adapter per artist (all linear layers)
- ▶ **Blend** two adapters in delta space (rank concatenation):

$$A_{\text{blend}} = \begin{bmatrix} \alpha A_1 \\ (1 - \alpha) A_2 \end{bmatrix}, \quad B_{\text{blend}} = \begin{bmatrix} B_1 & B_2 \end{bmatrix}$$

α = weight of artist 1; endpoints recover the pure adapters exactly. Now: did it actually learn the artist?

Evaluation is hard: generated lyrics have no label



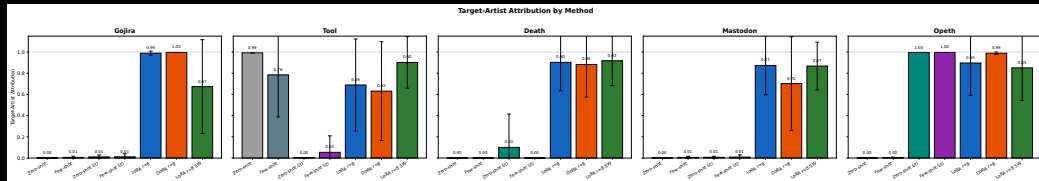
- ▶ No gold reference for a generated line $\Rightarrow$ train a **judge**
- ▶ RoBERTa-base 5-way attribution classifier, accuracy **0.84**, macro-F1 **0.83**
- ▶ Strong in-domain (Death f1 1.00)
- ▶ But it is *one* automatic judge — a **sanity check, not ground truth** (we'll see it can be gamed)

If a generation is attributed to its target artist with no artist name in the prompt, the style is in the *text* — as long as we don't over-trust a single number.

Three comparisons we run

- ▶ **LoRA vs DoRA** (weight-decomposed variant)
- ▶ **LoRA rank ablation** (r4 / r8 / r16)
- ▶ **Plain LoRA vs style-weighted (SW) LoRA** (TF-IDF-reweighted loss)

Adapters attribute; baselines collapse to a sink



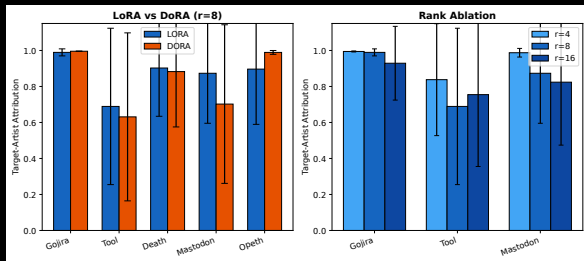
- ▶ **No “other” class:** any out-of-distribution text is forced onto the *nearest* real artist.
- ▶ **Not class imbalance:** the majority class is Mastodon (154), yet base → **Tool**, instruct → **Opeth** — different sinks, neither the majority.

The sink is a *measurement artifact*, not a class prior. Broadly, the judge still points the right way: adapters land on-target.

Plain LoRA r8

Target	Attribution
Gojira	0.989
Death	0.903
Opeth	0.896
Mastodon	0.873
Tool	0.689
base → Tool	≈0.99
instruct → Opeth	≈0.99

Ablation: LoRA vs DoRA & LoRA rank



Rank	Gojira	Tool	Mastodon
r4	.994	.838	.987
r8	.989	.689	.873
r16	.929	.755	.824

r16 consistently overfits; r4 matches/beats r8 on attribution but can lose on perplexity — rank choice depends on the metric.

What the adapters actually write

(plain LoRA r8 — one sample per artist)

Gojira

*I took a boat across the ocean
And I sail away, sailing home
On an open sea that is so deep*

Mastodon

*The storm is coming down
And sweeping through the trees
Cleansing spirit, dispelling dreams*

Death

*In the dark of night
The dead come forth from their
graves
A bloody ritual is being performed*

Tool

*My mind's on a different frequency
So I might not see eye to eye with
you
I hear voices calling me away*

Opeth

*I had a dream I was in the fall
That everything would die away
And when winter comes there is no
more*

Coherent, on-theme, no artist
name in the prompt. So is the
judge enough?

Beyond the judge: lexical metrics (and their limits)

- ▶ **Types / sample** — distinct artist-marker tokens used $\Rightarrow$ lexical *breadth*.
- ▶ **Occurrences / sample** — total marker hits $\Rightarrow$ inflated by *repetition*.
- ▶ **distinct-2** — fraction of unique bigrams $\Rightarrow$ global *degeneration* (looping).

Limitations — no metric is neutral:

- ▶ Token-level TF-IDF favors **concrete-noun** artists (Gojira: boat/star/forest) over **register/phrasal** (Tool) and **long-word** (Death, split by BPE into unreadable fragments).
- ▶ Occurrences *reward* a model that loops one marker; the judge can be gamed the same way.

So we read all of them together — and they don't agree.

How the judge scores them: higher score, worse lyrics

Plain Tool — score 0.003

*You know better than me
What's what and why
I might not be a prodigy
But I can write a rhyme*

Style-weighted Tool — score 0.996

*Yeah, yeah
Yeah, hey, hey
Hey, hey, hey, hey
Hey, hey, hey, hey (×10)*

	Gojira (plain→SW)	Tool (plain→SW)
Attribution	.989 → .674	.689 → .901
Types/sample	1.6 → 5.7 (3.6×)	2.5 → 3.3 (1.3×)
Occ./type	1.2 → 2.1	1.2 → 8.6

A degenerate “hey” loop *beats* a coherent verse. Tool’s gain is pure repetition (types flat), not breadth.

Four metrics, five artists — they disagree

Artist	Attribution (plain→SW)	Types/sample (plain→SW)	Occ./sample (plain→SW)	distinct-2 (plain→SW)
Gojira	.99 → .67	1.6 → 5.7	1.9 → 11.9	.96 → .71
Tool	.69 → .90	2.5 → 3.3	3.1 → 28.5	.88 → .74
Death	.90 → .92	1.7 → 1.5	2.9 → 7.2	.67 → .63
Mastodon	.87 → .87	2.0 → 3.3	2.7 → 4.9	.90 → .70
Opeth	.90 → .85	0.7 → 2.8	0.7 → 7.1	.92 → .71

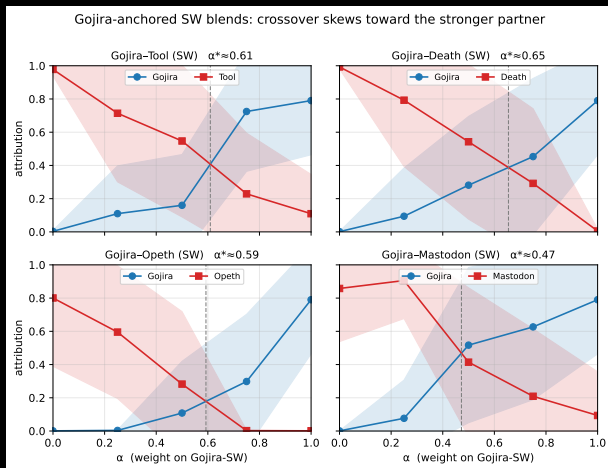
Breadth (types) rises for Gojira/Opeth/Mastodon but not Tool/Death; occurrences inflate everywhere; distinct-2 falls for all; attribution moves both ways. No single number captures style.

Perplexity — a clean diagonal, yet it disagrees too

adapter (r8)	held-out lyrics of				
	Gojira	Tool	Death	Mastodon	Opeth
base	45.9	18.6	30.4	32.3	57.7
<i>plain LoRA</i>					
Gojira	18.2	10.9	15.3	18.7	34.0
Tool	28.2	4.3	16.1	19.1	37.0
Death	70.3	24.6	4.5	49.5	105.6
Mastodon	40.4	14.6	23.0	8.1	54.4
Opeth	29.0	12.2	16.5	20.7	9.5
<i>style-weighted (SW)</i>					
Gojira	22.2	11.9	17.2	20.2	39.5
Tool	28.1	5.2	15.8	18.7	36.6
Death	49.8	19.1	6.6	36.8	71.3
Mastodon	35.5	13.5	20.7	11.6	47.1
Opeth	35.1	13.4	19.3	22.7	16.9

Both blocks specialize (own-artist = column min, **bold**); but SW *raises* every diagonal — it buys lexicon/attribution at a likelihood cost. Perplexity rewards what SW penalizes.

Blending: 50/50 weights $\neq$ 50/50 style



All four Gojira-SW blends cross over away from 0.5 ($\alpha^* = 0.47\text{--}0.65$): the partner usually wins a 50/50 mix. Tool alone shifted 0.17 (plain) $\rightarrow$ 0.61 (SW) — the loss, not the weights, decides.

Pure Gojira ($\alpha=1.0$) — Gojira 1.00

*No sense pretending to be innocent
Don't even try, no reason now
Kill this beast you call "reason"
Break the rules and break away the law*

*There is a vision coming alive in my mind
Like bright fireflies flying free
Awakening from the sleep I lay awake on the rock
I understand it all*

*Create your own reality!
You create yourself to become what you want to be
And fight for better
All alone against all of them*

The anchor: coherent, cosmic/creation imagery (realm / force / matter and energy / dimension).
Scored Gojira 1.00.

*Enter your own realm — open the door
Leave all illusions behind — step inside
Set a course straight — go along until the end
Open your eyes embrace!*

*Enter your own realm — open the door!
Enter your own realm! ! !*

*This force that runs through you creates the world
From matter and energy made — structures fall apart!
This force is running through me too — create another
Create — create — create — another dimension!*

Gojira + Tool ($\alpha=0.5$) — Tool 0.99

*I will rise and overcome
Leave this cage for better*

*Fight back, fight back
Don't you know I am alive?
Fight back, fight back
Don't you know I am alive?*

*Rise above the walls of hatred
Learn to feel compassion
Break the chains that bind us
And understand why we must fight*

*Feel our wrath on your skin
Feel our wrath on your skin
Feel our wrath on your skin
Feel our wrath on your skin*

*Don't you know I am alive?
Don't you know I am alive?
Don't you know I am alive?
Don't you know I am alive?
Don't you know I am alive?
Don't you know I am alive?
Don't you know I am alive?
Don't you know I am alive?*

*Don't you know I am alive?
Don't you know I am alive?
Don't you know I am alive?
Don't you know I am alive?*

- ▶ **Gojira:** defiant uplift — rise above hatred / break the chains / compassion
- ▶ **Tool:** the hypnotic 12× mantra
- ▶ **Verdict:** Tool 0.99 — the loop, not imagery, wins

Gojira + Mastodon ($\alpha=0.5$) — Tool 0.58 (a third artist)

*I wonder if I'll ever learn to fly
If it is right there in the sky
If all the stars align and shine so bright
Can you see them, baby? Can you count them?*

*And you said, "It's good for your health"
Good for your health
Oh yeah, yeah, yeah
Yeah, yeah, yeah*

*Get high, get high, get high
Fly free, fly free, fly free
Get high, get high, get high
Fly free, fly free, fly free
High above this beautiful world
I can fly
Fly through these open skies*

*In the night, the moon shines brightly on me
I can feel her warmth come down to earth
Bright eyes and teeth and golden skin
She belongs to Mother Nature, the Earth
Don't be afraid of falling down
We all crawl from mud and bone
Stand up tall and straight
And ride that wave of life*

*Get high, get high, get high
Fly free, fly free, fly free
Get high, get high, get high
Fly free, fly free, fly free
Get high, get high, get high
Fly free, fly free, fly free*

- ▶ **Gojira:** earth/nature — Mother Nature / mud and bone / wave of life
- ▶ **Mastodon:** astral flight — fly free / high above / stars align
- ▶ **Verdict:** judge says *Tool* (0.58) — a third artist

Gojira + Opeth ($\alpha=0.5$) — Gojira 0.99 (anchor holds)

*There's no escape
I am trapped in a maze
My soul is ablaze
My heart is aching for what it knows
Lost in space, lost on purpose
Dream state I enter now*

*Open the door to another world
Slip inside and let my spirit fly
Take my hand and guide me there
Help me find my way
Into the night
Inside you can hear nothing but silence
And that voice crying out for love
Leave all behind, leave this old shell
Leave alone, come back soon*

*Give us peace, give us time
Set us free, release the spell
Enter your mind
Outside is wild, dangerous land
We fight within the walls of our minds
Stay with me awhile until dawn appears
You are one of a kind
Don't ever forget
Never part again
This is our destiny, always and forever more...
Until then we shall meet again in another realm
Until then we shall meet again in another realm
No matter how long and hard you try
In vain try to forget what came before
How could you do this to me?
I thought you cared, but now I know why*

- ▶ **Gojira:** realm/another-world — spirit fly / another realm / enter your mind
- ▶ **Opeth:** melancholic loss — heart aching / "voice crying out for love" / "I thought you cared"
- ▶ **Verdict:** anchor holds — Gojira 0.99

Gojira + Death ($\alpha=0.5$) — Mastodon 0.79 (a third artist)

*Enter the land of darkness
An ancient race now destroyed
Lost in time and space forever
Never to return*

*In the sky so clear you can see
The flames burning bright below
A giant dragon is breathing fire
It grows ever stronger with each blow*

*Feel the heat flowing through your veins
Fire at will, breathe and destroy
Dragon is approaching fast
Breath comes out as red hot steel
Leave a trail of death behind you*

*Descend from the skies
You've arrived
Dragons of doom*

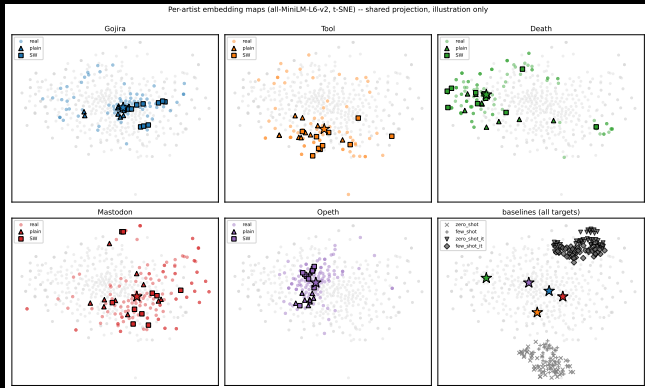
*Descend from the skies
You've arrived
Dragons of doom*

*Descend from the skies
You've arrived
Dragons of doom*

*Descend from the skies
You've arrived
Dragons of doom (×7)*

- ▶ **Gojira:** cosmic ruin — ancient race destroyed / lost in time and space
- ▶ **Death:** violence — breathe and destroy / red hot steel / trail of death
- ▶ **Verdict:** dragon/doom fantasy reads as *Mastodon* (0.79), loops “Dragons of doom” 7×

Independent instrument, same verdict — and the close



- ▶ Adapters land on each artist's real-song cluster; base/instruct sit in OOD islands.
- ▶ No single metric captures a “good” generation — judge, breadth, repetition disagree.
- ▶ The most trustworthy measure is **human evaluation**, which one person alone cannot supply at scale.

Style is not one number.